A Classical Approaches for Standard MDP

Dynamic programming provides classical approaches such as value iteration [6] and policy iteration [12], which are widely used for solving standard MDP problems as formulated in (1). In this work, we concentrate on value iteration, in which the key idea is to exploit the recursive structure of the Bellman optimality equation and iteratively approximate the optimal value function $V^*(s)$.

Specifically, for a given deterministic policy π, the Q-function $Q_\pi(s,a)$ satisfies the Bellman expectation equation

$$Q^\pi(s,a) = r(s,a) + \delta \mathbb{E}_{s'\sim P(\cdot|s,a)}[V^\pi(s')]. \tag{57}$$

In contrast, the optimal value function $V^*(s)$ is characterized by the Bellman optimality equation, given by

$$\begin{aligned} V^*(s) &= \max_a \left\{ r(s,a) + \delta \mathbb{E}_{s'\sim P(\cdot|s,a)}[V^*(s')] \right\} \\ &= \max_a Q^*(s,a). \end{aligned} \tag{58}$$

Let $B(\mathcal{S})$ denote the space of all bounded measurable functions $V : \mathcal{S} \to \mathbb{R}$, the Bellman operator $\mathcal{T} : B(\mathcal{S}) \to B(\mathcal{S})$ is defined as

$$[\mathcal{T}V](s) \triangleq \max_a \left\{ r(s,a) + \delta \mathbb{E}_{s'\sim P(\cdot|s,a)}[V(s')] \right\}. \tag{59}$$

Value iteration is simply the repeated application of the Bellman operator $\mathcal{T}$. For example, at each iteration k, the value function is updated by

$$V^{k+1}(s) = [\mathcal{T}V^k](s), \quad \forall s \in \mathcal{S}. \tag{60}$$

The convergence of the value iteration is guaranteed by the contraction property of the Bellman operator. Note that a Bellman operator $\mathcal{T}$ is called a contraction on a metric space $(d, B(\mathcal{S}))$ if there exists a constant $\delta \in (0,1)$ such that [3]

$$d([\mathcal{T}V](s), [\mathcal{T}U](s)) \le \delta d(V(s), U(s)), \quad \forall\, V(s), U(s) \in B(\mathcal{S}). \tag{61}$$

To establish the convergence of value iteration, we show in the following lemma that the Bellman operator $\mathcal{T}$ is a contraction mapping under the infinity-norm $\|\cdot\|_\infty$, which is defined as $\|V\|_\infty := \max_{s\in\mathcal{S}} |V(s)|$.

Lemma 7. *Bellman Operator $\mathcal{T}$ is a contraction mapping under infinity-norm $\|\cdot\|_\infty$, i.e., there exists $\delta \in (0,1)$ such that*

$$\|[\mathcal{T}V] - [\mathcal{T}U]\|_\infty \le \delta \|V - U\|_\infty. \tag{62}$$

Proof. For any state $s \in \mathcal{S}$, we have

$$\begin{aligned} \left|[\mathcal{T}V](s) - [\mathcal{T}U](s)\right| =& \Big| \max_{a\in\mathcal{A}} \left\{ r(s,a) + \delta \mathbb{E}_{s'\sim P(\cdot|s,a)}[V(s')] \right\} \\ & - \max_{a\in\mathcal{A}} \left\{ r(s,a) + \delta \mathbb{E}_{s'\sim P(\cdot|s,a)}[U(s')] \right\} \Big| \\ \overset{(a)}{\le}& \max_{a\in\mathcal{A}} \left| \delta \mathbb{E}_{s'\sim P(\cdot|s,a)}[V(s') - U(s')] \right| \\ \overset{(b)}{=}& \left| \delta \mathbb{E}_{s'\sim P(\cdot|s,a^*)}[V(s') - U(s')] \right| \\ \le& \delta \max_{s'\in\mathcal{S}} |V(s') - U(s')| \\ =& \delta \|V - U\|_\infty, \end{aligned} \tag{63}$$

where the inequality (a) follows from $|\max_a f(a) - \max_a g(a)| \le \max_a |f(a) - g(a)|$, and a^* in (b) denotes the optimal solution to the right-hand-side of (a).

Since the inequality in (63) holds for all state $s \in \mathcal{S}$, we can take the maximum over s to obtain

$$\|\mathcal{T}V - \mathcal{T}U\|_\infty \le \delta \|V - U\|_\infty. \tag{64}$$

Thus, $\mathcal{T}$ is a contraction mapping under the infinity-norm, which completes the proof. □

By Lemma 7 and Banach fixed-point theorem [3], the operator $\mathcal{T}$ admits a unique fixed point V^*, and value iteration converges to V^* from any initialization. Moreover, the rate of convergence is geometric as

$$\|V^k - V^*\|_\infty \le \delta^k \|V^0 - V^*\|_\infty, \tag{65}$$

implying that $\lim_{k\to\infty} V^k = V^*$. Once V^* is obtained, the optimal deterministic policy π^* can be recovered by acting greedily according to

$$\pi^*(s) = \arg\max_{a\in\mathcal{A}} \left\{ r(s,a) + \delta \mathbb{E}_{s'\sim P(\cdot|s,a)}[V^*(s')] \right\}, \tag{66}$$

which indicates that value iteration offers a simple yet rigorous method for computing the optimal policy.

B Classical Approaches for Fractional Programming

Consider a nonempty action set $\mathcal{A}$, a non-negative function $N(a)$, and a positive function $D(a)$, the fractional programming problem can be formulated as

$$\text{maximize}_a \quad \frac{N(a)}{D(a)}, \tag{67a}$$

$$\text{subject to} \quad a \in \mathcal{A}. \tag{67b}$$

The above fractional problem is generally non-convex, except for the special case such as numerator and denominator that can be reduced to a convex function. Unlike the convex problem which has a unique global optimal solution, the non-convex problem may have multiple local optimal solutions in contrast. This poses a challenge in solving such a fractional problem directly.

There exists a variety of techniques for solving FP problems. For the single-ratio case, Martos' algorithm [20] is a classical method for linear FP problems, while the Charnes-Cooper transform [7] provides an exact reformulation of such problems into linear programming. For non-linear FP problems, Dinkelbach's method [11] is a widely used iterative approach, which enjoys global convergence guarantees in the *concave-convex* setting, where the numerator $N(a)$ is concave in a and the denominator $D(a)$ is convex and positive. For the multiple-ratio case, [10] studied solving special max-min formulations, while the Quadratic transform proposed in [31] provides a general framework for addressing multiple-ratio FP problems. Given that Dinkelbach's method has been extensively applied to fractional MDP problems and the quadratic transform offers an efficient solution for handling multi-ratio structures, we restrict our review to these two approaches.

B.1 Dinkelbach's Transform

By introducing an auxiliary variable $\gamma \in \mathbb{R}$, Dinkelbach's method reformulates the single-ratio problem in (67) as

$$\Gamma(\gamma) \triangleq \underset{a}{\text{maximize}} \quad N(a) - \gamma D(a), \tag{68a}$$
$$\text{subject to} \quad a \in \mathcal{A}. \tag{68b}$$

The equivalence between the original FP and the transformed problem is established in the following lemma [11, Theorem].

Lemma 8 (Dinkelbach's Equivalence). *When γ^* is the optimum objective value of Problem (67), Problem (68) with $\gamma = \gamma^*$ is equivalent to Problem (67) and $\Gamma(\gamma^*) = 0$.*

Lemma 8 implies that if a^* is the optimal solution to Problem (67), then $N(a) - \gamma^* D(a) \leq 0$ and $N(a^*) - \gamma^* D(a^*) = 0$, indicating that a^* is also optimal to Problem (68). Specifically, the Dinkelbach's variable γ is iteratively updated by

$$\gamma_{\kappa+1} = \frac{N(a_\kappa)}{D(a_\kappa)}, \tag{69}$$

where κ is the iteration index. In particular, when Problem (67) satisfies the special *concave-convex* case, optimizing a for fixed γ in (68) is a convex problem, which guarantees the convergence to the global optimum. However, this method cannot be directly extended to multi-ratio fractional programming problems since the optimal solutions of the original fractional problem and its Dinkelbach reformulation coincide, but the optimal objective values differ.

B.2 Quadratic Transform

Shen *et al.* proposed the quadratic transform as a general technique for solving FP problems with a few desirable properties [31]. Introducing an auxiliary variable $\eta \in \mathbb{R}$, the transformed objective function $g(a, \eta)$ corresponding to (67) is required to satisfy:

- **C1:** *(Decoupling)* The new transformed objective has the form $g(a, \eta) = f(N(a))q_1(\eta) + h(D(a))q_2(\eta)$, where $f(\cdot)$, $h(\cdot)$, $q_1(\cdot)$, and $q_2(\cdot)$ are deterministic functions, thereby decoupling $N(a)$ and $D(a)$.
- **C2:** *(Equivalent Solution)* A solution a^* maximizes $N(a)/D(a)$ if and only if there exists some η^* such that (a^*, η^*) maximizes $g(a, \eta)$.
- **C3:** *(Equivalent Objective)* For any a, if $\eta^* = \arg\max_{\eta} g(a, \eta)$, then $g(a, \eta^*) = N(a)/D(a)$.
- **C4:** *(Concavity)* For fixed a, $g(a, \eta)$ is concave in η, i.e., $\partial^2 g(a, \eta)/\partial \eta^2 = -2D(a) \leq 0$.

Note that Dinkelbach's transform satisfies C1, C2 and C4, but not C3. C3 is the key distinction as the quadratic form preserves not only the optimal solution but also the optimal objective value, which is crucial for extending to multi-ratio FP problems. A novel quadratic reformulation that meets all four conditions is given by

$$\underset{a,\eta}{\text{maximize}} \quad g(a, \eta) \triangleq 2\eta\sqrt{N(a)} - \eta^2 D(a), \tag{70a}$$
$$\text{subject to} \quad a \in \mathcal{A}, \quad \eta \in \mathbb{R}. \tag{70b}$$

In the *concave-convex* case of Problem (67), Problem (70) becomes a convex problem, since for fixed η, $\sqrt{N(a)}$ is concave in a and increasing together with $D(a)$ is convex. Consequently, Problem (70) can be efficiently solved using standard convex optimization methods.

Different from Dinkelbach's transform, Shen's approach is applicable to multi-ratio fractional programming problems, as it preserves the equivalence of the original objectives. Let $\mathcal{M} = \{1, 2, \cdots, M\}$ denote the index set of numerator-denominator pairs. Given functions $\{N_m(a), D_m(a)\}_{m \in \mathcal{M}}$, the sum-of-ratio problem is formulated as

$$\underset{a}{\text{maximize}} \quad \sum_{m=1}^{M} \frac{N_m(a)}{D_m(a)}, \tag{71a}$$
$$\text{subject to} \quad a \in \mathcal{A}. \tag{71b}$$

By quadratic transformation, Problem (71) can be reformulated as

$$\underset{a,\boldsymbol{\eta}}{\text{maximize}} \quad \sum_{m=1}^{M} \left(2\eta_m \sqrt{N_m(a)} - \eta_m^2 D_m(a)\right), \tag{72a}$$
$$\text{subject to} \quad a \in \mathcal{A}, \quad \eta_m \in \mathbb{R}. \tag{72b}$$

Note that for fixed a, each auxiliary variable η_m admits a closed form solution $\eta_m^* = \sqrt{N_m(a)}/D_m(a)$, which yields the same objective value as in Problem (71). This equivalence guarantees the extension of the quadratic transform method to multi-ratio FP problems, in contrast to Dinkelbach's method, which fails to preserve the objective value due to the lack of Condition C3.

C Dinkelbach-based Method for Fractional MDPs

We observe from (OP-D) in (4) that the objective involves expectations over stochastic dynamics, where each decision influences not only the immediate reward but also the transition probabilities and subsequent returns. This dynamic coupling induces recursive dependencies that classical FP techniques, designed for static one-shot optimization, are unable to capture directly.

In addition, unlike standard MDP problems, both the reward function and the transition dynamics in fractional MDPs depend on fractional allocation weights. As a result, traditional dynamic programming methods such as value iteration cannot be applied directly, since the corresponding Bellman operator is no longer contractive. A generalized approach to solving fractional MDP problems is to reformulate it as a new equivalent standard one. In particular, we detail a value iteration algorithm based on Dinkelbach's method [26], which solves fractional MDPs directly and serves as a foundation for developing our proposed bi-value iteration method.

While [26] focuses on fractional MDPs under the average cost criterion, we extend this framework to the discounted cost case. Followed with (4), the fractional cost of any policy π is defined as

$$\gamma^\pi \triangleq \frac{\lim\limits_{N\to\infty} \mathbb{E}_s^\pi \left[\sum\limits_{t=0}^{N-1} \delta^t r_n(s_t, a_t)\right]}{\lim\limits_{N\to\infty} \mathbb{E}_s^\pi \left[\sum\limits_{t=0}^{N-1} \delta^t r_d(s_t, a_t)\right]}. \tag{73}$$

Based on Dinkelbach's method, for a given $\gamma^* \in \mathbb{R}$, Problem (4) is equivalent to a standard MDP by treating $r_n(s_t, a_t) - \gamma^* r_d(s_t, a_t)$ as the new reward function. Accordingly, the optimal policy for (4) is

obtained as

$$\pi^* = \arg\max_{\pi} \lim_{N\to\infty} \mathbb{E}^{\pi}\Big[\sum_{t=0}^{N-1} \delta^t (r_n(s_t,a_t) - \gamma^* r_d(s_t,a_t))\Big]. \quad (74)$$

We observe that the value iteration method via Dinkelbach's transformation requires two nested loops: an inner loop for value iteration and an outer loop for updating Dinkelbach's variable. Practical implementations in [33, 41] have applied this type of approach to constrained fractional SMDP problems. However, the quadratic transform technique [31] cannot be directly applied to (73) due to its non-linear combination of terms. In contrast to Dinkelbach's linear combination, which merges the numerator and denominator rewards into a single equivalent reward function, thereby converting the fractional MDP problem into a standard MDP one.

Despite this advantage, the Dinkelbach-based value iteration remains computationally expensive. Specifically, given the state space $\mathcal{S}$ and the action space $\mathcal{A}$, its overall computational cost scales with both the number of Dinkelbach updates and the number of value iteration steps, each of which requires evaluating all $|\mathcal{S}||\mathcal{A}|$ state–action pairs. Moreover, since Dinkelbach's transformation is applicable only to single-ratio objectives, such methods cannot be directly extended to sum-of-ratios formulations that arise in multi-user or multi-source fractional MDPs.

D Proof of Theorem 1

Consider a stationary deterministic policy $\pi\colon \mathcal{S} \to \mathcal{A}$ and let π^* be the optimal stationary deterministic policy for the fractional MDP in (4). We then define a *single-state deviation policy* π^a such that, at the initial state $s_0 = s$, the system takes action a, and thereafter follows the optimal policy π^* for all subsequent states and steps. Recall the definitions of the pair of value functions $V_N^{\pi}(s)$ and $V_D^{\pi}(s)$ in (7), as well as the pair of Q-functions $Q_N^{\pi}(s,a)$ and $Q_D^{\pi}(s,a)$ in (8), we have

$$V_N^{\pi^a}(s) = Q_N^*(s,a) \quad \text{and} \quad V_D^{\pi^a}(s) = Q_D^*(s,a). \quad (75)$$

Under the optimal policy π^*, if we take $a = \pi^*(s)$, then we have

$$V_N^*(s) = Q_N^*(s,\pi^*(s)) \quad \text{and} \quad V_D^*(s) = Q_D^*(s,\pi^*(s)). \quad (76)$$

Define the corresponding optimal fractional value as

$$\rho^*(s) = \frac{V_N^*(s)}{V_D^*(s)}, \quad \forall s \in \mathcal{S}. \quad (77)$$

By the optimality of π^* and $V_D^{\pi}(s) > 0$, for the alternative policy π^a, we must have

$$\frac{V_N^{\pi^a}(s)}{V_D^{\pi^a}(s)} \le \frac{V_N^*(s)}{V_D^*(s)} = \rho^*(s). \quad (78)$$

Since $V_D^{\pi^a}(s) = Q_D^*(s,a) > 0$ in (75), we can clear the denominator to obtain

$$Q_N^*(s,a) - \rho^*(s) Q_D^*(s,a) \le 0, \quad \forall a \in \mathcal{A}, \quad (79)$$

with equality attained when $a = \pi^*(s)$.

Using (76), it follows (79) that for any $a \in \mathcal{A}$, we have

$$\frac{Q_N^*(s,a)}{Q_D^*(s,a)} \le \frac{Q_N^*(s,\pi^*(s))}{Q_D^*(s,\pi^*(s))}. \quad (80)$$

Therefore, the optimal stationary deterministic policy can be characterized as

$$\pi^*(s) = \arg\max_{a\in\mathcal{A}} \frac{Q_N^*(s,a)}{Q_D^*(s,a)}, \quad \forall s \in \mathcal{S}. \quad (81)$$

The corresponding optimal pair of value functions is then given by

$$\boldsymbol{V}^*(s) = \boldsymbol{Q}^*(s,\pi^*(s)), \ \forall s \in \mathcal{S}. \quad (82)$$

This completes the proof.

E Proof of Lemma 5

Under Assumption 3, the numerator reward function $r_n(s,a)$ is concave in a, while the denominator reward $r_d(s,a)$ is convex. Formally, we have

$$\frac{\partial^2 r_n(s,a)}{\partial a^2} \le 0 \quad \text{and} \quad \frac{\partial^2 r_d(s,a)}{\partial a^2} \ge 0. \quad (83)$$

From the definition of the pair of Q-functions in Equation (8), we have

$$\boldsymbol{Q}^k(s,a) = \begin{pmatrix} Q_N^k(s,a) \\ Q_D^k(s,a) \end{pmatrix} = \begin{pmatrix} r_n(s,a) + \delta \mathbb{E}_{s'\sim P(\cdot|s,a)}[V_N^k(s')] \\ r_d(s,a) + \delta \mathbb{E}_{s'\sim P(\cdot|s,a)}[V_D^k(s')] \end{pmatrix}. \quad (84)$$

By part (ii) of Assumption 3, for each state $s \in \mathcal{S}$, the transition probability $P(s'|s,a)$ is affine in a for every $s' \in \mathcal{S}$. Consequently, for any bounded function $f : \mathcal{S} \to \mathbb{R}$, the expectation $\mathbb{E}_{s'\sim P(\cdot|s,a)}[f(s')]$ is also affine in a. Such affine transition models naturally arise in randomized or mixed-action control and scheduling problems, including the age of information settings considered in our experiments.

Under this assumption, the expectation term $\mathbb{E}_{s'\sim P(\cdot|s,a)}[\cdot]$ is an affine function of a and consequently does not alter the curvature of the Q-functions with respect to a. Therefore, since affine functions have zero second-order derivatives, the concavity of $Q_N^k(s,a)$ and convexity of $Q_D^k(s,a)$ are determined solely by the curvature of the immediate reward functions $r_n(s,a)$ and $r_d(s,a)$, respectively. Specifically, we have

$$\frac{\partial^2 Q_N^k(s,a)}{\partial a^2} = \frac{\partial^2 r_n(s,a)}{\partial a^2} \le 0 \text{ and } \frac{\partial^2 Q_D^k(s,a)}{\partial a^2} = \frac{\partial^2 r_d(s,a)}{\partial a^2} \ge 0, \quad (85)$$

which proves that $Q_N^k(s,a)$ is concave and $Q_D^k(s,a)$ is convex in a for all k and $s \in \mathcal{S}$.

F Proof of Lemma 4

Let $\boldsymbol{V} \in \mathcal{B}_{\bar{\rho}}$, so $\|V_c - V_c^*\|_\infty < \bar{\rho}$ for $c \in \{N, D\}$. For any (s,a), the Q-function perturbation satisfies

$$|Q_c^{\boldsymbol{V}}(s,a) - Q_c^*(s,a)| = \delta \left|\mathbb{E}_{s'}[V_c(s') - V_c^*(s')]\right| \le \delta\bar{\rho}, \quad (86)$$

where the equality uses $Q_c^{\boldsymbol{V}}(s,a) = r_c(s,a) + \delta\mathbb{E}_{s'}[V_c(s')]$ and $Q_c^*(s,a) = r_c(s,a) + \delta\mathbb{E}_{s'}[V_c^*(s')]$, and the inequality uses the triangle inequality and $P(\cdot|s,a)$ being a probability measure.

We bound the ratio perturbation at any (s,a). Define $\varepsilon'_N := Q_N^{\boldsymbol{V}}(s,a) - Q_N^*(s,a)$ and $\varepsilon'_D := Q_D^{\boldsymbol{V}}(s,a) - Q_D^*(s,a)$, so $|\varepsilon'_N|, |\varepsilon'_D| \le \delta\bar{\rho}$

by (86). Then

$$\frac{Q_N^{\boldsymbol{V}}(s,a)}{Q_D^{\boldsymbol{V}}(s,a)} - \frac{Q_N^*(s,a)}{Q_D^*(s,a)} = \frac{(Q_N^* + \varepsilon_N')Q_D^* - Q_N^*(Q_D^* + \varepsilon_D')}{Q_D^{\boldsymbol{V}} \cdot Q_D^*} = \frac{\varepsilon_N' Q_D^* - Q_N^* \varepsilon_D'}{Q_D^{\boldsymbol{V}} \cdot Q_D^*}. \quad (87)$$

Taking absolute values and applying the triangle inequality:

$$\left| \frac{Q_N^{\boldsymbol{V}}}{Q_D^{\boldsymbol{V}}} - \frac{Q_N^*}{Q_D^*} \right| \le \frac{|\varepsilon_N'| \cdot Q_D^* + Q_N^* \cdot |\varepsilon_D'|}{Q_D^{\boldsymbol{V}} \cdot Q_D^*} \le \frac{\delta\bar{\rho}(Q_N^* + Q_D^*)}{Q_D^{\boldsymbol{V}} \cdot Q_D^*}. \quad (88)$$

By Lemma 3, we obtain that $Q_N^*(s,a)+Q_D^*(s,a) \le R_\Sigma$ and $Q_D^*(s,a) \ge q_{\min}$. For the denominator, $Q_D^{\boldsymbol{V}}(s,a) = Q_D^*(s,a) + \varepsilon_D' \ge q_{\min} - \delta\bar{\rho}$. Since $\bar{\rho} < q_{\min}/\delta$, we have $q_{\min} - \delta\bar{\rho} > 0$. Substituting into (88), we have

$$\left| \frac{Q_N^{\boldsymbol{V}}(s,a)}{Q_D^{\boldsymbol{V}}(s,a)} - \frac{Q_N^*(s,a)}{Q_D^*(s,a)} \right| \le \frac{\delta\bar{\rho}R_\Sigma}{(q_{\min} - \delta\bar{\rho})\, q_{\min}}. \quad (89)$$

We now verify that this perturbation preserves the optimality of $\pi^*(s)$. For the optimal action $a^* = \pi^*(s)$ versus any suboptimal $a' \ne \pi^*(s)$, we have

$$\begin{aligned}
&\frac{Q_N^{\boldsymbol{V}}(s,a^*)}{Q_D^{\boldsymbol{V}}(s,a^*)} - \frac{Q_N^{\boldsymbol{V}}(s,a')}{Q_D^{\boldsymbol{V}}(s,a')} \\
&\ge \left(\frac{Q_N^*(s,a^*)}{Q_D^*(s,a^*)} - \frac{\delta\bar{\rho}R_\Sigma}{(q_{\min} - \delta\bar{\rho})q_{\min}} \right) - \left(\frac{Q_N^*(s,a')}{Q_D^*(s,a')} + \frac{\delta\bar{\rho}R_\Sigma}{(q_{\min} - \delta\bar{\rho})q_{\min}} \right) \\
&= \underbrace{\frac{Q_N^*(s,a^*)}{Q_D^*(s,a^*)} - \frac{Q_N^*(s,a')}{Q_D^*(s,a')}}_{\ge G_{\min}} - \frac{2\delta\bar{\rho}R_\Sigma}{(q_{\min} - \delta\bar{\rho})q_{\min}}.
\end{aligned} \quad (90)$$

We show the right-hand side is strictly positive. Substituting $\bar{\rho}$ from (20), $R_\Sigma = (r_{n,\max} + r_{d,\max})/(1-\delta)$, and $q_{\min} = r_{d,\min}/(1-\delta)$, we have

$$\begin{aligned}
&\frac{2\delta\bar{\rho}R_\Sigma}{(q_{\min} - \delta\bar{\rho})q_{\min}} = \frac{2\delta\bar{\rho}(r_{n,\max} + r_{d,\max})(1-\delta)}{r_{d,\min} \cdot \frac{2r_{d,\min}(r_{n,\max}+r_{d,\max})}{2(r_{n,\max}+r_{d,\max})+G_{\min}\cdot r_{d,\min}}} \\
&= \frac{\delta\bar{\rho}(1-\delta)\left[2(r_{n,\max} + r_{d,\max}) + G_{\min} \cdot r_{d,\min}\right]}{r_{d,\min}^2} \\
&= \frac{G_{\min} \cdot r_{d,\min}^2}{2(r_{n,\max} + r_{d,\max}) + G_{\min} \cdot r_{d,\min}} \cdot \frac{2(r_{n,\max} + r_{d,\max}) + G_{\min} \cdot r_{d,\min}}{r_{d,\min}^2} \\
&= G_{\min}.
\end{aligned} \quad (91)$$

Hence the perturbation term in (90) equals $G_{\min}$ exactly when $\|\boldsymbol{V} - \boldsymbol{V}^*\|_\infty = \bar{\rho}$. Since $\mathcal{B}_{\bar{\rho}}$ is an open ball ($\|\boldsymbol{V} - \boldsymbol{V}^*\|_\infty < \bar{\rho}$), the perturbation is strictly less than $G_{\min}$, ensuring

$$\frac{Q_N^{\boldsymbol{V}}(s,a^*)}{Q_D^{\boldsymbol{V}}(s,a^*)} - \frac{Q_N^{\boldsymbol{V}}(s,a')}{Q_D^{\boldsymbol{V}}(s,a')} > 0 \quad \forall\, a' \ne \pi^*(s). \quad (92)$$

Therefore $\pi_g(\boldsymbol{V})(s) = \pi^*(s)$ for all $s \in \mathcal{S}$.

Let $\boldsymbol{W} := \mathcal{T}\boldsymbol{V}$. By $\pi_g(\boldsymbol{V}) = \pi^*$, $\mathcal{T}$ reduces to the linear Bellman backup under π^*, given by

$$W_c(s) = r_c(s, \pi^*(s)) + \delta\mathbb{E}_{s' \sim P(\cdot|s,\pi^*(s))}[V_c(s')],\ c \in \{N, D\}. \quad (93)$$

Since $V_c^*(s) = r_c(s, \pi^*(s)) + \delta\mathbb{E}_{s'}[V_c^*(s')]$, we have

$$W_c(s) - V_c^*(s) = \delta\mathbb{E}_{s'}[V_c(s') - V_c^*(s')]. \quad (94)$$

Therefore $|W_c(s) - V_c^*(s)| \le \delta\|V_c - V_c^*\|_\infty < \delta\bar{\rho} < \bar{\rho}$, confirming the distance condition. For the positivity condition, using $\frac{r_{d,\min}}{1-\delta} \le V_D^\pi(s) \le \frac{r_{d,\max}}{1-\delta}$, we have

$$\begin{aligned}
W_D(s) &= r_d(s, \pi^*(s)) + \delta\mathbb{E}_{s'}[V_D(s')] \\
&\ge r_{d,\min} + \delta(V_D^*(s') - \bar{\rho}) \\
&\ge r_{d,\min} + \delta(q_{\min} - \bar{\rho}) = q_{\min} - \delta\bar{\rho} > 0,
\end{aligned} \quad (95)$$

where the last inequality uses $\bar{\rho} < q_{\min}/\delta$. Hence $\boldsymbol{W} \in \mathcal{B}_{\bar{\rho}}$.

Therefore, we establish the linear contraction as

$$\begin{aligned}
\|\mathcal{T}\boldsymbol{V} - \boldsymbol{V}^*\|_\infty &= \max_s \max_{c \in \{N,D\}} \left|\delta\mathbb{E}_{s'}[V_c(s') - V_c^*(s')]\right| \\
&\le \delta \max_c \|V_c - V_c^*\|_\infty = \delta\|\boldsymbol{V} - \boldsymbol{V}^*\|_\infty,
\end{aligned} \quad (96)$$

where the inequality uses P^{π^*} being a stochastic matrix. This completes the proof.

G Proof of Theorem 3

For any $k \ge 0$ and any (s,a), the fractional Bellman backup yields $|Q_c^{k+1}(s,a) - Q_c^k(s,a)| = \delta|\mathbb{E}_{s'}[V_c^{k+1}(s') - V_c^k(s')]| \le \delta e^k$ for $c \in \{N, D\}$. Using the algebraic identity $\frac{a_1}{b_1} - \frac{a_2}{b_2} = \frac{(a_1-a_2)b_2-(b_1-b_2)a_2}{b_1 b_2}$ together with $Q_D^k, Q_D^{k+1} \ge q_{\min}$ and $Q_N^k + Q_D^k \le R_\Sigma$, we obtain

$$\left| \frac{Q_N^{k+1}(s,a)}{Q_D^{k+1}(s,a)} - \frac{Q_N^k(s,a)}{Q_D^k(s,a)} \right| \le \frac{\delta\, e^k\, R_\Sigma}{q_{\min}^2}. \quad (97)$$

Next, we prove by induction on $j \ge 0$ that $\mathbf{H}(j)\colon \pi^{K+i+1} = \pi^{K+1}$ for all $0 \le i \le j$, and $e^{K+i} \le \delta^i e^K$ for all $0 \le i \le j$.

For the base case $j = 0$, $\mathbf{H}(0)$ holds by definition. Assume $\mathbf{H}(j)$ holds for all $0 \le i \le j$. Because the policy is fixed at π^{K+1} from iteration $K+1$ through $K+j$, the iteration reduces to the linear Bellman backup under a fixed policy, so

$$V_c^{K+j+1}(s) - V_c^{K+j}(s) = \delta\,\mathbb{E}_{s'}[V_c^{K+j}(s') - V_c^{K+j-1}(s')], \quad (98)$$

which yields $e^{K+j+1} \le \delta\, e^{K+j} \le \delta^{j+1} e^K$. Applying (97) cumulatively between the ratios at $\boldsymbol{V}^K$ and $\boldsymbol{V}^{K+j+1}$, and using that the greedy gap G^{K+j+1} is a minimum over at most a $\frac{2\delta e^{K+i} R_\Sigma}{q_{\min}^2}$ perturbation per step,

$$G^{K+j+1} \ge G^K - \frac{2\delta R_\Sigma}{q_{\min}^2} \sum_{i=0}^{j} e^{K+i} \ge G^K - \frac{2\delta e^K R_\Sigma}{q_{\min}^2 (1-\delta)} > 0, \quad (99)$$

where the last inequality is exactly the self-certification criterion (25). The strict positivity of the gap at iteration $K+j+1$ implies $\pi^{K+j+2} = \pi^{K+j+1} = \pi^{K+1}$, completing the induction.

Under the lock $\pi^k = \pi^{K+1}$ for all $k \ge K+1$, the bi-value iteration degenerates to the linear Bellman backup $\boldsymbol{V}^{k+1} = \boldsymbol{r}^{\pi^{K+1}} + \delta P^{\pi^{K+1}} \boldsymbol{V}^k$, which contracts with factor δ and thus converges to $\boldsymbol{V}^{\pi^{K+1}}$, the exact value pair of π^{K+1}. Letting $j \to \infty$ in (99) shows $G^\infty \ge G^K - \frac{2\delta e^K R_\Sigma}{q_{\min}^2 (1-\delta)} > 0$, which means π^{K+1} is greedy with respect to $\boldsymbol{V}^{\pi^{K+1}}$. Hence $\boldsymbol{V}^{\pi^{K+1}}$ is a fixed point of the fractional Bellman operator $\mathcal{T}$. By Theorem 1, the fractional Bellman equation admits a unique fixed point corresponding to π^*, so $\pi^{K+1} = \pi^*$ and $\boldsymbol{V}^{\pi^{K+1}} = \boldsymbol{V}^*$.

Since $e^{K+j} \le \delta^j e^K$, telescoping gives, for any $k \ge K+1$,

$$\|\boldsymbol{V}^k - \boldsymbol{V}^*\|_\infty \le \sum_{j=0}^{\infty} \|\boldsymbol{V}^{k+j+1} - \boldsymbol{V}^{k+j}\|_\infty \le \sum_{j=0}^{\infty} \delta^{k-K-1+j} e^K = \frac{\delta^{k-K} e^K}{1-\delta}, \tag{100}$$

which establishes (26) and completes the proof.

H Proof of Theorem 4

The proof follows the invariant-set argument of Lemma 4, extended to the product space. For each $m \in \mathcal{M}$ and $\boldsymbol{V}_{\mathcal{M}} \in \mathcal{B}^{(M)}_{\bar{\rho}^{(M)}}$, by the same computation as (89), the per-ratio perturbation is given by

$$\left| \frac{Q^{\boldsymbol{V}}_{N,m}(s,a)}{Q^{\boldsymbol{V}}_{D,m}(s,a)} - \frac{Q^*_{N,m}(s,a)}{Q^*_{D,m}(s,a)} \right| \le \frac{\delta \bar{\rho}^{(M)} \cdot R_\Sigma}{(q_{\min} - \delta \bar{\rho}^{(M)}) \cdot q_{\min}}. \tag{101}$$

By the triangle inequality over M ratio terms, we obtain the sum-of-ratios perturbation as

$$\left| \sum_{m=1}^{M} \frac{Q^{\boldsymbol{V}}_{N,m}}{Q^{\boldsymbol{V}}_{D,m}} - \sum_{m=1}^{M} \frac{Q^*_{N,m}}{Q^*_{D,m}} \right| \le \frac{M \delta \bar{\rho}^{(M)} \cdot R_\Sigma}{(q_{\min} - \delta \bar{\rho}^{(M)}) \cdot q_{\min}}. \tag{102}$$

For $\pi_g(\boldsymbol{V}_{\mathcal{M}}) = \pi^*$, we need (102)$< G^{(M)}_{\min}/2$. Substituting $\bar{\rho}^{(M)}$ from (37) and following identical algebraic steps as in the proof of Lemma 4, the gap is preserved since $\mathcal{B}^{(M)}_{\bar{\rho}^{(M)}}$ is open. Within $\mathcal{B}^{(M)}_{\bar{\rho}^{(M)}}$, the multi-ratio Bellman operator reduces to component-wise linear backups under π^*, i.e.,

$$(\mathcal{T}\boldsymbol{V}_{\mathcal{M}})_m = r^{\pi^*}_m + \delta P^{\pi^*} \boldsymbol{V}_m, \quad m = 1, \ldots, M. \tag{103}$$

Each component contracts at rate δ. Taking the maximum over m yields $\max_m \|\boldsymbol{V}^{k+1}_m - \boldsymbol{V}^*_m\|_\infty \le \delta \max_m \|\boldsymbol{V}^k_m - \boldsymbol{V}^*_m\|_\infty$. Therefore, the linear convergence follows by induction, which completes the proof.

I Dinkelbach-based Value Iteration for AoI Minimization

By Dinkelbach's reformulation, Problem (44) is equivalent to solving

$$\underset{\pi}{\text{minimize}} \quad \mathbb{E}[A(Y,Z,Y')] - \gamma \mathbb{E}[Y+Z], \tag{104a}$$

$$\text{subject to} \quad 0 \le Z \le Z_{\max}, \tag{104b}$$

where γ denotes the Dinkelbach variable associated with the ratio transformation. Then, the corresponding reward function is defined as

$$r(\boldsymbol{s}, \boldsymbol{a}) = \frac{1}{2}(y+z)^2 + (y+z)y' - \gamma(y+z), \tag{105}$$

where the $\boldsymbol{a} = (z, n)$ consists of the waiting time z and the server choice n. Given a fixed γ, the value function is iterated by

$$V^{k+1}(\boldsymbol{s}) = \min_{\boldsymbol{a}} \Big[\frac{1}{2}(y+z)^2 + (y+z)y' - \gamma(y+z) + \mathbb{E}_{\boldsymbol{s}' \sim P(\cdot|\boldsymbol{s},\boldsymbol{a})}[V^k(\boldsymbol{s}')] \Big]. \tag{106}$$

To obtain the optimal value function, we define the Q-function as

$$Q^k(\boldsymbol{s}, \boldsymbol{a}) = \frac{1}{2}(y+z)^2 + (y+z)y' - \gamma(y+z) + \mathbb{E}_{\boldsymbol{s}' \sim P(\cdot|\boldsymbol{s},\boldsymbol{a})}[V^k(\boldsymbol{s}')]. \tag{107}$$

Since the set of servers is finite, the optimal server index can be determined by evaluating all possible $n = \{1, 2, \cdots, N\}$. For each server n, we optimize the waiting time z_n by minimizing $Q^k(\boldsymbol{s}, (z_n, n))$ over $z \in [0, Z_{\max}]$ via the gradient descent method. In particular, the derivative of $Q^k(\boldsymbol{s}, (z_n, n))$ with the respect to z is given by

$$\frac{\partial Q^k(\boldsymbol{s}, (z_n, n))}{\partial z} = y + z + y' - \gamma. \tag{108}$$

Hence, the action $\boldsymbol{a}^k = (z^k, n^k)$ at k-th iteration is then determined as

$$n^k = \arg\min_n Q^k(\boldsymbol{s}, (z^k_n, n)) \quad \text{and} \quad z^k = z_{n^k}. \tag{109}$$

The Dinkelbach parameter γ is updated in the outer iteration as

$$\gamma_\kappa = \frac{\mathbb{E}\left[\frac{1}{2}(y+z)^2 + (y+z)y'\right]}{\mathbb{E}\left[y+z\right]}, \quad \forall \kappa. \tag{110}$$

As $\kappa \to \infty$, γ_κ converges to the optimal γ^*.